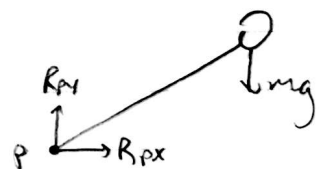
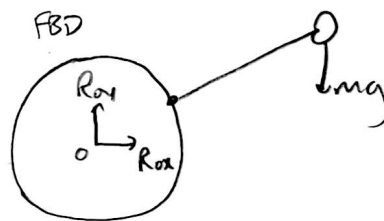
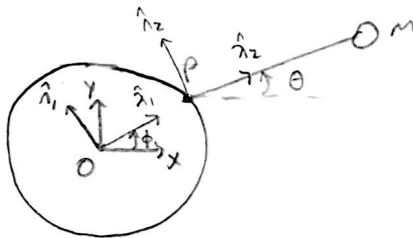
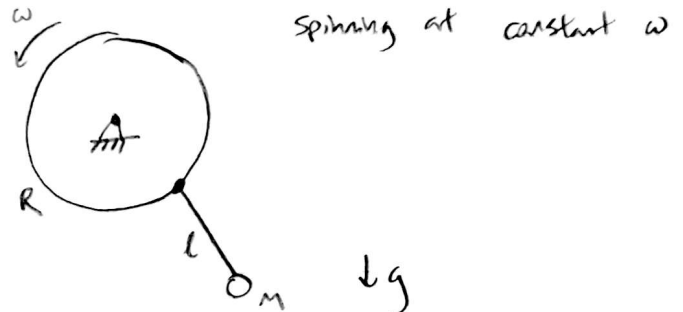


DYNAMICS Q-Exam 1 Page 30 #3

Find EOM of mass M



$$\begin{aligned}\vec{r}_{M/O} &= R \hat{n}_1 + l \hat{n}_2 \\ \vec{v}_{M/O} &= R\omega \hat{n}_1 + l\dot{\theta} \hat{n}_2 \\ \vec{a}_{M/O} &= -R\omega^2 \hat{n}_1 + l\ddot{\theta} \hat{n}_2 - l\dot{\theta}^2 \hat{n}_2\end{aligned}$$

	$\hat{i}$	$\hat{j}$
$\hat{n}_1$	$\cos\phi$	$\sin\phi$
$\hat{n}_2$	$-\sin\phi$	$\cos\phi$
$\hat{e}_1$	$\cos\theta$	$\sin\theta$
$\hat{e}_2$	$-\sin\theta$	$\cos\theta$

AMB / P of PM

$$\sum \vec{M}_{P/P} = \vec{H}_{P/P}$$

$$\vec{r}_{M/P} \times -mg \hat{j} = \vec{r}_{M/P} \times M \vec{a}_{M/O}$$

$$l \hat{n}_2 \times -mg \hat{j} = l \hat{n}_2 \times M (-R\omega^2 \hat{n}_1 + l\dot{\theta} \hat{n}_2 - l\dot{\theta}^2 \hat{n}_2)$$

$$l \cos\theta \hat{i} \times -mg \hat{j} = (l \cos\theta \hat{i} + l \sin\theta \hat{j}) \times M [(-R\omega^2 \cos\phi - l\dot{\theta} \sin\theta - l\dot{\theta}^2 \cos\theta) \hat{i} + (-R\omega^2 \sin\phi + l\dot{\theta} \cos\theta - l\dot{\theta}^2 \sin\theta) \hat{j}]$$

$$\begin{aligned}-gl \cos\theta &= l \cos\theta (-R\omega^2 \sin\phi + l\dot{\theta} \cos\theta - l\dot{\theta}^2 \sin\theta) + l \sin\theta (R\omega^2 \cos\phi + l\dot{\theta} \sin\theta + l\dot{\theta}^2 \cos\theta) \\ &= l \cos\theta (-R\omega^2 \sin\phi - l\dot{\theta}^2 \sin\theta) + l \sin\theta (R\omega^2 \cos\phi + l\dot{\theta}^2 \cos\theta) + l^2 \ddot{\theta}\end{aligned}$$

$$\ddot{\theta} = \frac{1}{l} [\cos\theta (-g + R\omega^2 \sin\phi + l\dot{\theta}^2 \sin\theta) - \sin\theta (R\omega^2 \cos\phi + l\dot{\theta}^2 \cos\theta)]$$

$$\boxed{\ddot{\theta} = \frac{1}{l} (-g \cos\theta + R\omega^2 \sin\phi \cos\theta - R\omega^2 \cos\phi \sin\theta)}$$

What if $\omega = 0$? System should behave as a simple pendulum

$$\ddot{\theta} = \frac{1}{l} (-g \cos\theta + R\omega^2 \sin\phi \cos\theta - R\omega^2 \cos\phi \sin\theta)$$

$\ddot{\theta} = \frac{1}{l} (-g \cos\theta)$ ← solution for simple pendulum in this coordinate system

```
% Donald Feng df266
% Rotating single pendulum solution
% Uses rpendrhs.m
% User specifies initial conditions and

clear functions
clear all
close all

% System parameters
p.R = 1;
p.l = 2;
p.w = 0.5;
p.g = 1;
ttot=40; %duration of animation in seconds

% Initial conditions
phi0 = 0;
th0 = 0;
thdot0 = 0;

z0 = [phi0 th0 thdot0];

% Solve and plot
framespersec=40; %if view is not speeded or slowed in fourbaranimate
tspan=linspace(0,ttot,ttot*framespersec);

options=odeset('abstol',1e-6,'reltol',1e-6);
[t z] = ode45('rpendrhs',[tspan],z0,options, p);

% Animation of pendulum
clf

n=length(t); dur=t(end);

% Reference before translation and rotation
bar1ref = p.R*[0 1;0 0];
bar2ref = p.l*[0 1;0 0];

hinge=plot(0,0, 'marker','.', 'markersize',[50], ...
           'color','black');hold on;

axis('equal')
maxL = p.R+p.l;
axis([-maxL maxL -maxL maxL]);
title('trajectory of rotating pendulum')

bar1pic=plot(bar1ref(1,:),bar1ref(2,:), ...
             'linewidth', 3, 'color','green');
bar2pic=plot(bar2ref(1,:)+1,bar1ref(2,:), ... % +1 for init disp
             'linewidth', 3, 'color','red');
```

```

set(bar1pic, 'linewidth',7) % fatten bar 1 a bit

pause(1)% Hold still for a second before starting

for i=1:n
    pause(.01) %Use to control the speed of the animation

    th1=z(i,1); th2=z(i,2);
    R1=[cos(th1) -sin(th1) ; sin(th1) cos(th1)]; % one rotation matrix
    R2=[cos(th2) -sin(th2) ; sin(th2) cos(th2)]; % another rotation
    matrix

    bar1=R1*bar1ref;
    bar2=R2*bar2ref; bar2offset=bar1(:,2);

    set(bar1pic, 'xdata', bar1(1,:), 'ydata', bar1(2,:));
    set(bar2pic, 'xdata', bar2(1,:)+bar2offset(1), ...
        'ydata', bar2(2,:)+bar2offset(2) );
    barx = bar2(1,2)+bar2offset(1);
    bary = bar2(2,2)+bar2offset(2);
    if i>1
        barxends(i-1) = barx;
        baryends(i-1) = bary;
    end
    if i>1
        plot(barxends, baryends, '--b')
    end
end

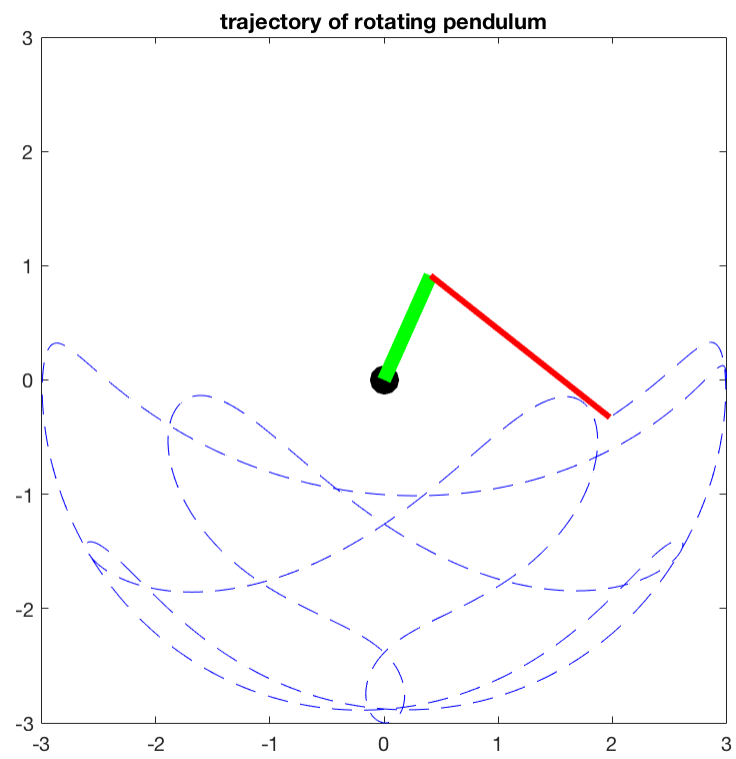
function zdot = rpendrhs(t,z,flag,p)

R = p.R;
l = p.l;
g = p.g;
w = p.w;
phi = z(1);
th = z(2);
thdot = z(3);

thddot = (-g*cos(th)+R*w^2*sin(phi)*cos(th)-R*w^2*cos(phi)*sin(th))/l;

zdot = [w thdot thddot]';
end

```



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